

Employee Attrition Analysis Summary

Employee attrition is a major challenge for organizations because replacing employees requires significant time, cost, and effort. In this project, employee data was analyzed to identify patterns associated with employee turnover and to build a prediction system that can help HR teams identify employees who may be at risk of leaving.

The analysis revealed that the overall attrition rate was approximately 16.12%. The Sales department experienced the highest attrition rate, followed by Human Resources. Among job roles, Sales Representatives showed the highest turnover rate, indicating that retention efforts should be focused on this group.

Income was found to have an impact on attrition, as employees who left the company had a lower average monthly income compared to those who stayed. Work-Life Balance also played an important role. Employees with the lowest work-life balance ratings were significantly more likely to leave the organization.

The predictive models were evaluated and Logistic Regression provided the best overall performance for identifying employees at risk of attrition. Key factors influencing employee turnover included overtime work, frequent business travel, job role, promotion history, and overall work experience.

Based on these findings, HR teams should focus on reducing excessive overtime, improving employee work-life balance, and providing targeted retention programs for employees in high-risk roles such as Sales Representatives and Laboratory Technicians. The model can support HR decision-making, but it should be used alongside human judgment because personal and organizational factors cannot be fully captured through historical data alone.